Salimli Ayzek

Data Scientist. 1st-year master's student, SPbSTU, ICCS. Mathematics and Computer Science, Machine Learning and Artificial Intelligence.

thebloomsbridge - Work

GitHub

Hugging Face

ORCID

ResearchGate

eLibrary

Saint-Petersburg

Версия на русском 🇷🇺

Download resume (ENG)

Download this portfolio

About

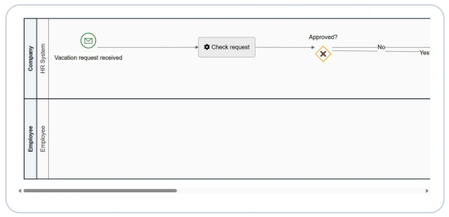
- Salimli Ayzek - master's student at SPbSTU (ICCS), majoring in Mathematics and Computer Science: Machine Learning and Artificial Intelligence. I have 1 year and 5 months of experience at The Blooms Bridge: AI systems integration (RAG, computer vision, etc.) and backend development. I primarily position myself as a Data Scientist and work with the ML cycle: data preprocessing and transformation, data analysis, feature selection, model training, hyperparameter tuning, result interpretation, data labeling, LLM training, model building (PyTorch, TensorFlow), deployment (LangChain + LangGraph), and computer vision tasks (YOLO, ResNet, MobileNet, ViT, U-Net, SegFormer, YOLO-seg, YOLO-pose, MediaPipe, OpenCV).
- My professional interests include computer vision, statistical and machine learning algorithms, neural networks, functional programming, optimization methods, statistics, and discrete mathematics.
- Outside work: travel, philosophy (especially how cognitive thinking and foundational ideas connect with computer science), football, and neue deutsche harte.

Projects

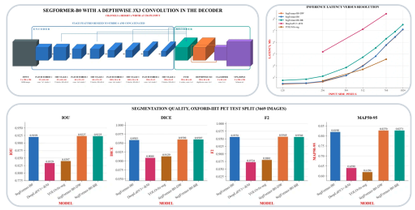
ML and AI

Object detection on dynamic RoI based on segmentation with tracking (weights available)

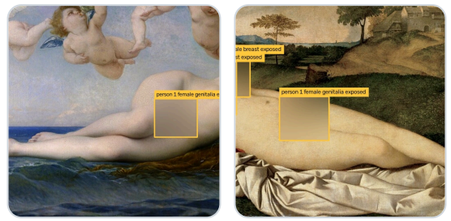
Text-to-BPMN (generating BPMN diagrams from text)



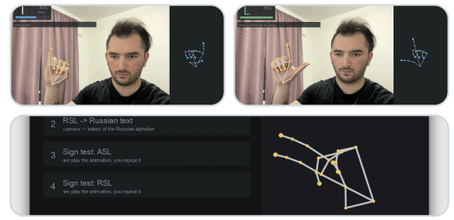
SegFormer depthwise enhancement (weights on HuggingFace)



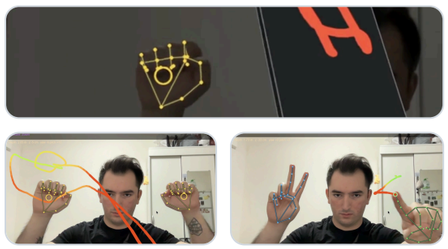
Real-time censorship API (for online meetings)



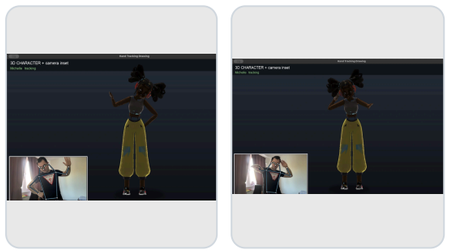
ASL, RSL sign languages to text (text to sign) translator



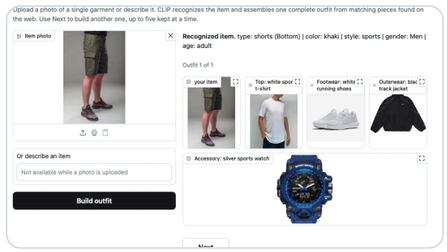
Hand tracking drawing (with 3D)



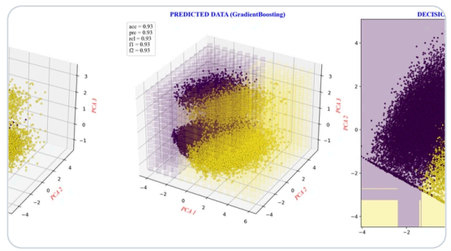
Pose driven 3D characters



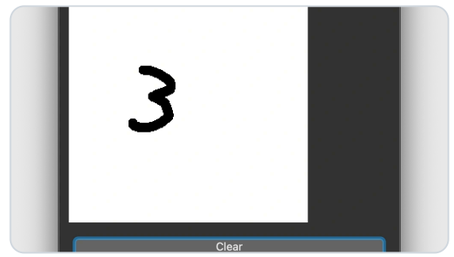
Vision fashion recommender



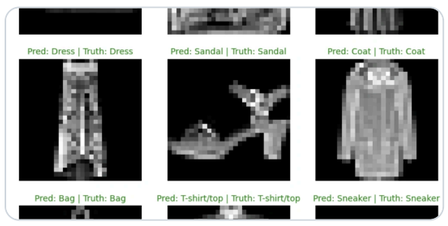
Heart disease classifier



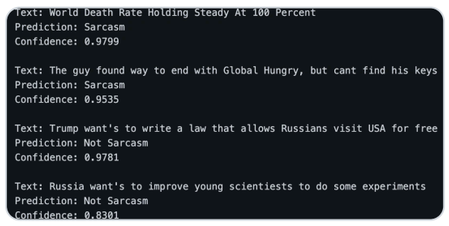
Qt MNIST recognizer



Fashion-MNIST classification with TinyVGG

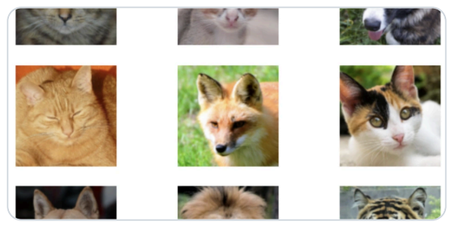


Sarcasmic text classification (transformer NN)

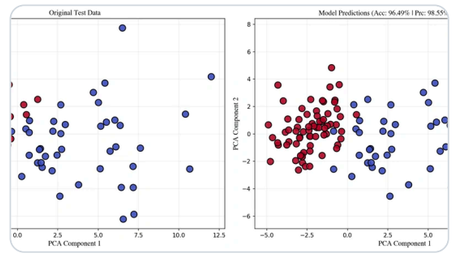


Leafs health classification (GoogLeNet)

Classification of cats, dogs and wild animals (CNN)



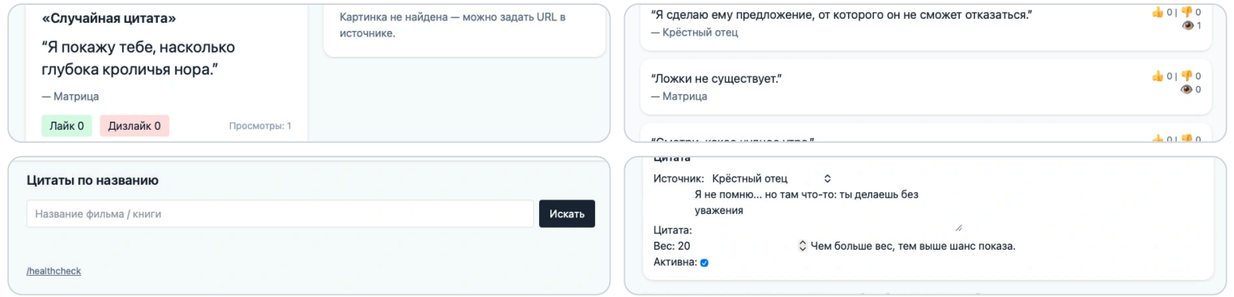
Breast cancer classification



Web

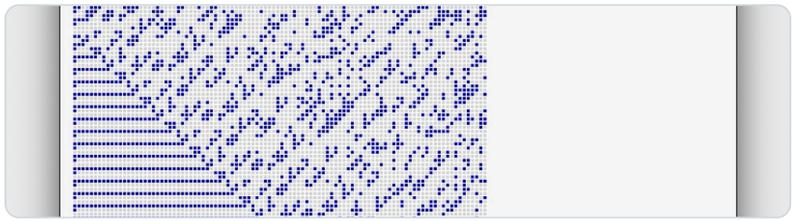
Data encryption and decryption

Quotes from movies and cartoons

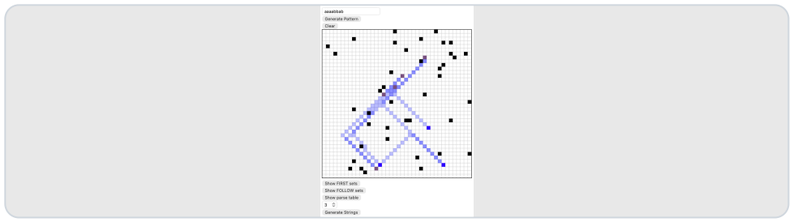


Soft

Game of life on CA



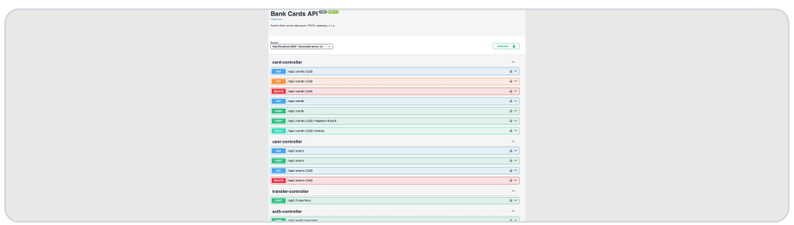
LL(1) grammar with BFS semantics on Haskell



Spring based TG-Bot: Trip planner

AyzekSounds: Open source MP3 player for iOS

Card management service



Math

Forecasting and comparison of demographic indicators of Russia and Japan

Regression analysis of the consequences of the reduction in gas export

Bachelor's degree works (SPbSTU, 02.03.01_1)

ПОЛИТЕХ
Институт компьютерных наук и кибербезопасности

Master's degree works (SPbSTU, 02.04.01_1)

ПОЛИТЕХ
Институт компьютерных наук и кибербезопасности

Education

School
Complete secondary

Azerbaijan - Baku

School №53

Mathematics focus

Bachelor

Russia - Saint Petersburg

Peter the Great St. Petersburg Polytechnic University

Institute of Computer Science and Cybersecurity

Mathematics and Computer Science - AI Systems and Supercomputer Technologies

Graduate School of Artificial Intelligence Technologies

Master's degree
1st year

Russia - Saint Petersburg

Peter the Great St. Petersburg Polytechnic University

Institute of Computer Science and Cybersecurity

Mathematics and Computer Science - Machine Learning and Artificial Intelligence

Graduate School of Artificial Intelligence Technologies

Experience

- **The Blooms Bridge** - **ML Engineer** (20.07.2024 - present)
  - Integration of AI systems, including computer vision and agentic workflows, into the project using RAG, YOLO, YOLO-seg, SegFormer, OpenCV, LangChain, and LangGraph
  - Development of a project dashboard and algorithms for dynamic vehicle routing problem (dVRP) tasks
  - Research and training of semantic and instance segmentation models for real-time segmentation tasks
  - Development of an automated system for reconciling customer orders with the items collected by couriers
  - Development of a RAG system that provides recommendations based on reports
  - Clustering of users by their orders and geographic areas
  - Development of a zero-shot recommendation system for finding similar products
  - Forecasting the number of products required for procurement
  - Statistical and time-series analysis using ARIMA, SARIMA, and SARIMAX
  - Development of a dynamic route optimization algorithm accounting for the courier's mode of transport, road events, product characteristics, and weather conditions
  - Research into replacing the ESA attention mechanism with state-space models (SSMs) in ViT-style segmenters
- **Private tutoring** - **Mathematics Tutor** (02.04.2023 - 03.05.2024)
  - Preparation for state exams in mathematics
  - Linear algebra for applicants to technical degree programs

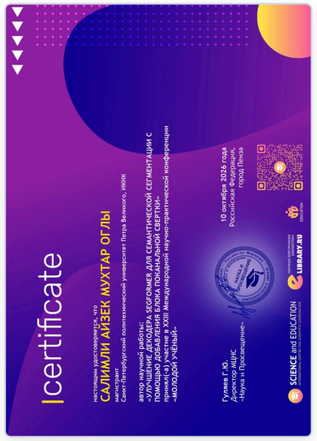
Articles

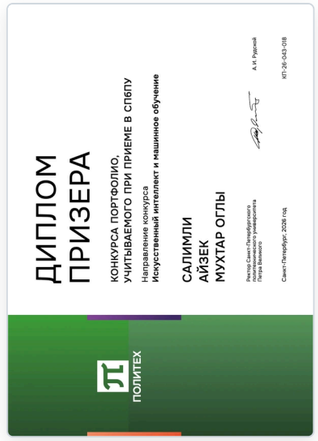
- Improving the SegFormer decoder for semantic segmentation by adding a depthwise convolution block (page 52)[2]
- Construction of the Region of Interest based on segmentation[1]
- Computer vision for offside detection in football

[1] **- Recognized as one of the best articles.**

[2] **- 1st place in the "Technical Sciences" section.**

Diplomas and Certificates





Skills

Category	Technologies / Libraries
Languages	Python · Haskell
Data Science, ML & CV	PyTorch · TensorFlow · NumPy · Pandas · PySpark · Scikit-learn · YOLO · OpenCV · PIL
MLOps	MLflow · Hugging Face · ZenML
Visualization	Matplotlib · Plotly · Seaborn
BI & Analytics	Tableau · Excel
Scientific	SciPy · Statsmodels · Cirq
DBMS & Message Brokers	PostgreSQL · MongoDB · RabbitMQ
DevOps (basics)	Docker · Kubernetes · Jenkins · Terraform · Ansible
Cloud (basics)	OpenStack Heat · Yandex Cloud · Firebase · Render
Mathematics	• Statistical Learning • Optimization • Linear Algebra • Calculus • Probability & Statistics • Graph Theory • Discrete Math • Quantum Computing

Learning Hub (on progress)

- *Computer Science*
- *Mathematics*

Contacts

- Telegram
- GitHub profile
- Gitea profile